CSE 400 Lecture Scribe

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Create a high-quality, exam-focused lecture scribe for CSE 400 (Lectures 15–17) using only the provided slides and prescribed textbook sections. Do not include any external knowledge, intuition, or added explanations.

The scribe must include all definitions and notation exactly as introduced; all assumptions, conditions, and constraints; all theorems, lemmas, and formal results written completely; all proofs and derivations presented step-by-step with no skipped steps, preserving the original logical structure; all algorithms/procedures (if any) in clear stepwise form; and all worked examples only if covered in the lecture, with full intermediate steps.

Ensure the output is well-structured and exam-ready, using clear headings, bullet points for definitions/assumptions, and numbered steps for proofs. Use proper mathematical notation and keep it concise but complete.

Strictly do not add intuition, interpretations, alternative methods, extra examples, or omit any steps. The final scribe should be compact, rigorous, and fully self-contained for direct exam revision.

Here is your **exam-focused, fully self-contained lecture scribe (Lectures 15–17)** strictly based on the provided slides:

CSE 400 – Lecture Scribe (L15–L17)

Joint Random Variables and Joint Distributions

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1. Motivation: Joint Random Variables

- In many situations, we consider **two (or more) random variables simultaneously**.
- Examples (as listed in slides):
  - Height and weight
  - IQ and birthweight
  - Exercise frequency and heart disease
  - Air pollution and illness rate
  - Number of friends and age
- Joint distributions allow:
  - Computation of probabilities involving multiple variables
  - Understanding relationships between variables
- If variables are not independent:
  - Dependence is analyzed using **covariance and correlation**

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2. Discrete Case – Joint PMF

2.1 Setup

- Let:
  - $X = \{x_1, x_2, \ldots, x_n\}$
  - $Y = \{y_1, y_2, \ldots, y_m\}$
- Then:
  - (X, Y) takes values in:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

2.2 Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

- Represents the probability of the **joint outcome occurring simultaneously**

2.3 Joint PMF Table Representation

- Organized as a table:

$X \backslash Y$	y_1	y_2	$\cdots$	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	$\cdots$	$p(x_1, y_m)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	$\cdots$	$p(x_2, y_m)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	$\cdots$	$p(x_n, y_m)$

Key Properties

1.

0 \le p(x_i, y_j) \le 1

]

2.

\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1

]

2.4 Example: Rolling Two Dice

Experiment

- Roll two fair dice

Define

- X: value on first die
- T: total (sum) of both dice

Joint Probability

- Each outcome (i, j) has probability:

$$\frac{1}{36}$$

- Construct joint PMF table of (X, T) (shown on slide table).

Observations

- Entries are either:
  - 0 or $\frac{1}{36}$
- X and T are **not independent**

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3. Continuous Case – Joint PDF

3.1 Analogy with Discrete Case

- Discrete → Continuous:
  - Values → Intervals
  - PMF → PDF
  - Sums → Double integrals

3.2 Support (Domain)

- If:
  - $X \in [a, b]$
  - $Y \in [c, d]$
- Then:

$$(X, Y) \in [a, b] \times [c, d]$$

3.3 Definition (Joint PDF)

A function $f(x, y)$ is a joint PDF if:

$$P((X, Y) \text{ in small area } dx\, dy) \approx f(x, y)\, dx\, dy$$

3.4 Key Properties

1.

f(x,y) \ge 0

]

2.

\int_c^d \int_a^b f(x,y),dx,dy = 1

]

- Note:
  - $f(x, y)$ may be **greater than 1** (density, not probability)

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4. Worked Example (Continuous Case)

Given

$$f(x, y) = 4xy, \quad 0 \le x \le 1,\ 0 \le y \le 1$$

- Domain: unit square $[0, 1]^2$

Task 1: Check Validity (Normalization)

Step-by-step:

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

$$= \int_0^1 \left(\int_0^1 4xy\, dx \right) dy$$

$$= \int_0^1 4y \left(\int_0^1 x\, dx \right) dy$$

$$= \int_0^1 4y \left[\frac{x^2}{2} \right]_0^1 dy$$

$$= \int_0^1 4y \cdot \frac{1}{2}\, dy$$

$$= \int_0^1 2y\,dy$$

$$= \left[y^2\right]_0^1 = 1$$

Conclusion:

- $f(x, y) \geq 0$ and total probability = 1
- Hence **valid joint PDF**

Task 2: Compute $P(A)$

Event:

$$A = \{X < 0.5,\ Y > 0.5\}$$

Step-by-step:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx$$

$$= \int_0^{0.5} \left(\int_{0.5}^1 4xy\,dy \right) dx$$

$$= \int_0^{0.5} 4x \left[\frac{y^2}{2} \right]_{0.5}^1 dx$$

$$= \int_0^{0.5} 4x \cdot \frac{1 - 0.25}{2} dx$$

$$= \int_0^{0.5} 4x \cdot \frac{0.75}{2} dx$$

$$= \int_0^{0.5} 2x(0.75) dx$$

$$= \int_0^{0.5} 1.5x\,dx$$

$$= \left[\frac{1.5x^2}{2} \right]_0^{0.5} = \frac{3}{16}$$

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5. Joint Cumulative Distribution Function (CDF)

5.1 Notation

- Event:

$$X \leq x,\ Y \leq y \equiv \{X \leq x \text{ and } Y \leq y\}$$

5.2 Definition

$$F(x, y) = P(X \leq x,\ Y \leq y)$$

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6. Joint CDF – Continuous Case

If X, Y have joint PDF $f(x, y)$ over $[a, b] \times [c, d]$:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering Joint PDF

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

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7. Joint CDF – Discrete Case

If joint PMF $p(x_i, y_j)$:

$$F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation

- Sum all probabilities:
  - Rows with $x_i \le x$
  - Columns with $y_j \le y$

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8. Properties of Joint CDF

1. **Non-decreasing:**

$$F(x, y) \text{ does not decrease if } x \text{ or } y \text{ increases}$$

2. **Lower-left limit:**

$$\lim_{(x,y) \to (-\infty, -\infty)} F(x, y) = 0$$

3. **Upper-right limit:**

$$\lim_{(x,y) \to (\infty, \infty)} F(x, y) = 1$$

End of Scribe

This is **complete, compact, and exam-ready**, strictly preserving all definitions, formulas, steps, and structure from the provided lecture slides with no added interpretation.